JAN-C-590

12 MAY 1948

JOINT ARMY-NAVY SPECIFICATION

CELLULOSE-ACETATE-BUTYRATE
(For Use in Protective Ointment)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Chemical Corps Tentative Specification
197-54-184A
15 Mar. 1944

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-B-107—Boxes, Wood, Wirebound (Overseas Type)
JAN-P-108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-P-138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.
JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. ARMY SPECIFICATIONS

19-8 —Liner, Shipping Container, Waterproof Barrier.
100-2—Standard Specification for Marking Shipments by Contractors. [1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material. [2]

FEDERAL SPECIFICATIONS

NN-B-591 —Boxes; Wood-Cleated-Fiberboard.
NN-B-601 —Boxes; Wood-Cleated-Plywood, For Domestic Shipment.
NN-B-621 —Boxes; Wood, Nailed and Lock-Corner.
NN-B-631 —Boxes; Wood, Wirebound, For Domestic Shipment.
LLL-B-631—Boxes; Fiber, Corrugated (For Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (For Domestic Shipment).

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

B. GRADE

B–1. This specification covers one grade of cellulose acetate butyrate, as hereinafter specified. (See par. H–1.)

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—

C–1a. The material shall be newly prepared stock. No reworked stock from other processes shall be used.

C–1b. The material shall be manufactured from purified, prime quality cellulose.

C–2. *Workmanship.*—All work shall be done in a neat and workmanlike manner.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Acidity.*—The acidity shall not exceed 0.01 percent, calculated as acetic acid.

E–2. *Moisture content.*—The material shall contain not more than 1.0 percent moisture at the time of packing.

E–3. *Solubility.*—The material shall provide clear, smooth solutions when dissolved "as received" to the extent of 12 percent, by weight, in dry ethylene dichloride. It shall be more than 98 percent soluble in this solvent.

E–4. *Stability.*—The material shall not become darker than a light tan when heated for 2 hours at 105°C.

E–5. *Esterification.*—

E–5a. *Acetyl content.*—The acetyl content shall be 13 ± 2 percent (15.4 to 21 percent combined acetic acid).

E–5b. *Butyryl content.*—The butyryl content shall be 39 ± 2 percent (45.8 to 50.8 percent combined butyric acid).

E–5c. *Saponification number.*—The saponification number shall be the total saponification ester calculated as percent acetyl and shall be 37.0 ± 0.5 percent (51 to 52.4 percent combined acetic acid).

E–6. *Copper.*—The copper content of the material shall not exceed 5 parts per million.

E–7. *Iron.*—The iron content of the material shall not exceed 15 parts per million.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be

[3] Applicable only to Navy purchases.

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afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—Unless otherwise specified in the contract or order, a lot shall consist of a single manufacturing batch of cellulose acetate butyrate. (See par. H–3.)

F–3. *Sampling.*—(See par. H–4.)

F–3a. *For moisture test.*—A 100-gm. representative sample shall be taken from each lot at the place of manufacture, immediately prior to packing, for the moisture test prescribed in paragraph F–4c.

F–3b. *For other tests.*—Each lot of cellulose acetate butyrate shall be sampled as follows: Take a ½-pound representative sample from each of 10 containers selected at random from the lot. If there are fewer than 10 containers in the lot, sample all the containers. Mix the samples thoroughly to form a composite sample representative of the lot. Place the composite sample in a clean, dry container, seal tightly, and label the container to identify the sample with the lot represented.

F–4. *Tests.*—

F–4a. *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents as used in the test, and corrections shall be applied when significant.

F–4b. *Acidity.*—Dissolve approximately 10.0 gm. of the material in a mixture of 200 ml. of neutral acetone and 20 ml. of water. When completely dissolved, add 50 ml. of water and shake well. Add 3 drops of methyl red indicator and titrate to a lemon-yellow end point with 0.01*N* sodium hydroxide solution. Calculate the percentage acetic acid as follows:

$$\text{Percentage acetic acid} = \frac{6.0AN}{W}$$

where:

A = ml. sodium hydroxide solution used in titration
N = normality of sodium hydroxide solution
W = weight of sample.

F–4c. *Moisture content.*—Weigh approximately 10.00 gm. of the sample in a tared porcelain dish. Heat the dish and contents in an oven at 105° ± 5°C. for 1 hour, cool in a desiccator, and weigh. Calculate the moisture content as follows:

$$\text{Percentage moisture content} = \frac{100A}{W}$$

where:

A = loss in weight on heating
W = weight of sample.

F–4d. *Solubility.*—Place 24 gm. of the material and 176 gm. of dry ethylene dichloride in a bottle and tumble overnight. The resulting solution shall be clear and smooth.

F–4e. *Esterification.*—The procedure consists in determining the total acyl content of the ester by alkaline hydrolysis of one

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sample in alcoholic solution, and then in hydrolyzing a larger sample with alkali in aqueous solution, treating the reaction mixture with phosphoric acid, distilling out the volatile acids, and determining the molar ratios of the acids in the distillate by titration before and after its partition with butyl acetate. From these data the composition of the ester can be calculated by means of simultaneous equations.[4]

F–4e(1). *Determination of total acyl.*—Grind approximately 6.5 gm. of the sample in a Wiley mill (or approved equivalent) and dry for 1 hour at approximately 50°C. in a vacuum oven. Place duplicate 1-gm. samples in 250-ml. Erlenmeyer flasks and add 40 ml. of 75-percent (by volume) ethyl alcohol to each flask. Place heavy lead collars around the flasks, stopper them loosely and heat them in a water bath at approximately 55°C. for 30 minutes to swell the fibers. Add 40.00 ml. of standard 0.5*N* alkali to each flask, stopper the flasks loosely, and heat them at approximately 55°C. for 15 minutes. Rinse down the sides of the flasks with a little alcohol, stopper the fiasks tightly, and allow them to stand at least 2 days at room temperature. Titrate the excess alkali with standard 0.5*N* hydrochloric acid, using phenolphthalein as the indicator. After reaching a temporary end point, allow the flask to stand at least 30 minutes so the alkali can diffuse from the fibers. Make further additions of standard acid until all the free alkali has been neutralized. Calculate the total acyl to apparent acetyl as follows:[5]

$$\text{Percentage apparent acetyl} = 43(AM - BN)$$

where:

A = ml. standard alkali added to each flask
M = normality of standard alkali
B = ml. standard hydrochloric acid used in back titre
N = normality of standard hydrochloric acid.

F–4e(2). *Isolation of mixed acids.*—Place duplicate 2.0-gm. samples (which need not be weighed accurately) in 125-ml. Erlenmeyer flasks. To each flask add 80.00 ml. of standard 0.5*N* alkali. Place lead collars around the flasks, and then place the flasks, loosely stoppered, in a water bath at 40° ± 1°C. When they have been heated to 40°C., stoppered tightly, and leave at 40° ± 1°C for at least 48 hours.

To each flask add, from a burette, a quantity of standard 1*N* phosphoric acid exactly equivalent to the 30.00 ml. of standard alkali originally added, calculated as follows:

$$\text{Milliliters phosphoric acid required} = \frac{80.00N}{M}$$

where:

N = normality of standard alkali used
M = normality of standard phosphoric acid.

[4] Further details on the esterification method may be found in *Analytical Procedure for the Determination of Acetyl, Propionyl, and Butyryl in Cellulose Esters,* by C. J. Malm and G. S. Nadeau, U. S. Patent 2,069,892, assigned to Eastman Kodak Company.

[5] In case a large number of specimens are being analyzed, a graph should be constructed by plotting percentage apparent acetyl against milliliters standard hydrochloric acid, then the percentage apparent acetyl in each sample can be read from the graph.

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The phosphoric acid forms monosodium phosphate and liberates the organic acids from their sodium salts. This addition of phosphoric acid to the alkaline reaction mixture should be made just prior to distillation.

The distillation apparatus recommended consists of a flask as shown on figure 1, a condenser, and a receiver. The flask has no capillary inlet tube, such as is cutomary in a vacuum distillation flask, since it was found to be unnecessary in the distillation of dilute acid from an aqueous reaction mixture under the conditions imposed. The distillation flask outlet is connected to a vertical Allihn condenser, the lower end of which extends within 3 inches of the bottom of a 500-ml., side-arm distilling flask which serves as a receiver. The latter is placed in a 1-gallon pail and surrounded with ice and salt. Transfer the hydrolyzed 2-gm. sample, after adding phosphoric acid and shaking the mixture, to the distillation flask through the 4-mm. stopcock.

Raise a pan of hot water, preferably heated nearly to boiling, so that the water surrounds the distilling flask, and evacuate the apparatus with a water vacuum pump. After the distillation has proceeded for approximately 15 minutes, it will be found that most of the liquid has distilled from the flask. Heat the water bath with a free flame until the flask becomes dry. (No harm results if it is heated indefinitely beyond this point.) Allow 25 ml. of water to enter the flask through the stopcock and continue the distillation until the flask becomes dry a second time. Rinse the condenser and remove the receiver from its ice-salt bath. (A distillation requires approximately 45 minutes.)

In this operation it is not necessary to work with quantitative accuracy at all stages, but it is necessary to obtain water solution of the acids in the same ratios as they occur in the esters. The volume of the distillate and rinsings is usually 200 to 225 ml., which in most cases automatically adjusts the acidity of the distillate to 0.06 to 0.08*N*, the range desired for subsequent extractions.

F–4e(3). *Determination of the molar ratios of the acids.*—The procedure is based on the distribution ratios of the various acids between water and n-butyl acetate.[6] The ratios are determined for the individual acids using samples of known high purity. From these values and the distribution ratios of the mixtures the molar ratios can be calculated, as described below.

Titrate a 25-ml. portion of the distillate against 0.05*N* sodium hydroxide solution using phenolphthalein indicator. Designate the volume of alkali consumed as M. This titration, as well as all subsequent ones, should be made in a current of carbon-dioxide-free air in a suitable titration flask. Shake 30 ml. of the distillate in a small separatory funnel with 15 ml. of n-butyl acetate. (These volumes should be measured accurately using pipettes or burettes.) After shaking the mixture thoroughly for at least 30 seconds and

[6] Further details of the method are described by C. H. Workman in *Iowa State Journal of Science*, Vol. 4, July 1930, and in *Journal of Industrial and Engineering Chemistry*, Analytical Edition, Vol. 3, p. 264, 1931.

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allowing at least 10 minutes for the layers to separate, draw off the lower (aqueous) layer. Measure out a 25-ml. portion and titrate against the 0.05*N* alkali. Designate the volume of alkali consumed as M_1. In the same way, extract a second 30-ml. portion of the distillate, using 60 ml. of the extractant. Again titrate 25 ml. of the aqueous layer. Designate the volume of alkali consumed as M_2. Calculate the following ratios:

$$\frac{100M_1}{M} = K_1 \qquad \frac{100M_2}{M} = K_2$$

K_1 and K_2 are termed the percentage partition coefficients of the acids in the distillate. They represent the percentage of acid remaining in the aqueous phase after extraction.

In a similar manner, determine the distribution ratios for acetic, propionic and butyric acids. Dilute a sample of each acid, of tested purity, with water to provide 200 to 300 ml. of approximately 0.07*N* solution. Titrate 25-ml. portions and extract and titrate 30-ml. portions as described above. Determine the following partition coefficients, leaving the values as decimal fractions rather than as percentages, thus:

$$\frac{M_1}{M} = k_1$$

k_1 = Partition coefficient of acetic acid with 15 ml. n-butyl acetate
k_2 = partition coefficient of propionic acid with 15 ml. n-butyl acetate
k_3 = partition coefficient of butyric acid with 15 ml. n-butyl actetate
k_4 = partition coefficient of acetic acid with 60 ml. n-butyl acetate
k_5 = partition coefficient of propionic acid with 60 ml. n-butyl acetate
k_6 = partition coefficient of butyric acid with 60 ml. n-butyl acetate

These constants should be checked occasionally and must be redetermined for each new supply of butyl acetate. Blanks should be run on the butyl acetate, for it develops acidity on standing. All measurements should be made with accurate pipettes or burettes, and extreme care and cleanliness should be observed during the entire operation. The accuracy of the procedure should be checked occasionally using a solution of a known mixture of the three acids.

F–4e(4). *Calculations.*—In order to evaluate three unknowns it is necessary to obtain three simultaneous algebraic equations involving the three unknown quantities. In the case of a ternary acid mixture, the sum of the mol percentages of the acids present represents the total acidity, or 100 percent. If A, P, and B represent mol percentages of acetic, propionic, and butyric acids, respectively, the first equation becomes:

(1) $A + P + B = 100$

The second and third equations are:

(2) $Ak_1 + Pk_2 + Bk_3 = K_1$

(3) $Ak_4 + Pk_5 + Bk_6 = K_2$

Note that k_1, k_2, k_3, k_4, k_5, and k_6 are known and refer to the pure individual acids, whereas K_1 and K_2 refer to the ternary mixture.

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By solving these equations for P, the following expressions may be derived:

(4) $$P = \frac{\frac{K_1 - 100k_1}{k_3 - k_1} - \frac{K_2 - 100k_4}{k_6 - k_4}}{\frac{k_2 - k_1}{k_3 - k_1} - \frac{k_5 - k_4}{k_6 - k_4}}$$

(5) $$B = \frac{K_2 - 100k_4 - P(k_5 - k_4)}{k_6 - k_4}$$

Typical values for the partition coefficient of the acids are:

$k_1 = 0.820$ $k_4 = 0.521$
$k_2 = 0.546$ $k_5 = 0.221$
$k_3 = 0.255$ $k_6 = 0.0759$

When these values are substituted in the preceding equations, they simplify to the following forms:

(6) $P = 9.35(K_1 - 82.0) - 11.88(K_2 - 52.1)$
(7) $B = 2.25(52.1 - K_2) - 0.674\ P.$
(8) $A = 100 - P - B.$[7]

The mol percentages of acid are converted into weight percent acyl of ester as follows:

Weight percent acetyl = A × percentage apparent acetyl
Weight percent propionyl = P × percentage apparent acetyl × $\frac{57}{43}$
Weight percent butyryl = B × percentage apparent acetyl × $\frac{71}{43}$

In case an acetopropionate or an acetobutyrate, definitely known to be such, is being analyzed, the procedure may be shortened by determining only one distribution ratio for a distillate. In this case the calculation of the result is greatly simplified.

F–4e(5). The percentage composition may be converted to an equivalent basis by means of the following formulas:

$$x = \frac{162.1a}{4300 - 42.39b - 42a}$$

$$y = \frac{0.605bx}{a}$$

Hydroxyl content in equivalents = $3 - (x + y)$

where:

x = acetyl content in equivalents
y = butyryl content in equivalents
a = weight percent acetyl
b = weight percent butyryl

When the weight percent propionyl is less than 0.5, it may be added to the butyryl weight percent in making the above calculations without affecting the results appreciably.

[7] The above constants and simplified equations are typical, but the exact values should be determined by each analyst using his own reagents and the best pure acids available. As previously mentioned, these constants must be checked from time to time, and particularly when new reagent solutions are put into use.

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F–4f. *Copper and iron content.*—

F–4f(1). *Preparation of material for iron and copper determinations.*—Weigh 10.0 gm. of the material into a porcelain crucible. Moisten with 1.0 ml. of 1:1 sulfuric acid. Ignite carefully and ash to a barely visible red heat in a muffle furnace. Allow to cool and determine the copper and iron content in the ash.

F–4f(2). *Copper content.*—Add 5 ml. of concentrated hydrochloric acid, one drop of bromine water, and 1 ml. of 1:10 sulfuric acid to the ash in the crucible. Digest to effect complete solution and evaporate to near dryness to remove bromine and hydrochloric acid. Dissolve the residue in a few ml. of water and transfer to a 60-ml. separatory funnel, controlling the final volume to 10 ml., indicated by a mark on the funnel. Add 5.0 ml. of dithizone reagent (see par. F–4f(4)a) and shake vigorously for 10 minutes. If the color of the dithizone layer turns from green to violet rather quickly, add 10.0 ml. more of the reagent and continue shaking. Copper, present to the extent of 2 parts per million in a 10-gm. sample, will change 5.0 ml. of dithizone reagent from green to red. A slight excess is necessary to assure that the color is between a green and a red. Discharge the reagent layer into a comparison tube or into a colorimeter cup. Reserve the aqueous layer for the iron test described in paragraph F–4f(3). Prepare standards containing known concentrations of copper by shaking measured volumes of a standard copper solution (see par. F–4f(4)b) made up to 9 ml. with water, and 1.0 ml. of 1:10 sulfuric acid, with the same volume of dithizone reagent as was used in the test. Compare the color developed by the sample in visual comparison tubes or in a colorimeter with those of the standards. Calculate the copper content in terms of parts of copper per million as follows:

$$\text{Parts per million of copper} = \frac{1{,}000{,}000A}{W}$$

where:

A = weight of copper found
W = weight of sample.

Run a blank on all reagents including the distilled water which may contain traces of copper.

F–4f(3). *Iron content.*—Add 1 ml. of carbon tetrachloride to the aqueous layer in the separatory funnel (see par. F–4f(2)), shake and discard the carbon tetrachloride layer. Add 10 ml. of 5-percent potassium thiocyanate solution and discharge the contents of the funnel into comparison tubes or into a colorimeter cup. Prepare standards containing known concentrations of iron by mixing measured volumes of a standard iron solution (see par. F–4f(4)c), 1.0 ml. of 1:10 sulfuric acid and 10 ml. of 5-percent potassium thiocyanate solution, made up to a total volume of 20 ml. Compare the color developed by the sample in visual comparison tubes or in a colorimeter with those of the freshly pre-

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pared standards. Calculate the iron content in terms of parts of iron per million, as follows:

$$\text{Parts per million of iron} = \frac{1{,}000{,}000A}{W}$$

where:

A = weight of iron found
W = weight of sample.

Run a blank on all reagents including the distilled water which may contain traces of iron.

F–4f(4). *Special reagents.*—

F–4f(4)a. *Dithizone reagent.*—Weigh out 20.0 mg. of dithizone (reagent grade) and dissolve in a small quantity of carbon tetrachloride. Filter into a 500-ml. volumetric flask, wash the filter with carbon tetrachloride, and dilute to the mark with carbon tetrachloride. Transfer the solution to a brown, glass-stoppered bottle.

F–4f(4)b. *Standard copper solution.*—Dissolve approximately 1.3 gm. of cupric chloride ($CuCl_2.2H_2O$) in a liter of water containing 10 ml. of concentrated hydrochloric acid. Determine the copper content of this solution by a standard method. From a burette run a sufficient quantity of this solution into a liter volumetric flask to make a concentration of 10.00 mg. of copper in one liter. Add 10 ml. of concentrated hydrochloric acid and dilute to the mark with water. One ml. of this solution is equivalent to 10 micrograms of copper; 5 ml. of this solution are equivalent to 5 parts per million of copper in a 10-gm. sample.

F–4f(4)c. *Standard iron solution.*—Dissolve approximately 4.0 gm. of ferric chloride ($FeCl_3.6H_2O$) in a liter of water containing 10 ml. of concentrated hydrochloric acid. Determine the exact ferric iron content of this solution by a standard method. From a burette run a sufficient quantity of this solution into a liter volumetric flask to make a concentration of 20.00 mg. of iron in one liter. Add 10 ml. of concentrated hydrochloric acid and dilute to the mark with water. One ml. of this solution is equivalent to 20 micrograms of iron; 7.5 ml. of this solution are equivalent to 15 parts per million of iron in a 10-gm. sample.

F–4f(5). *Rejections.*—If the representative sample fails to pass the tests for iron and copper, the lot shall be resampled and retested for iron and copper. If the new sample fails to pass these tests, the entire lot shall be rejected.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging and packing.*—

G–1a. *For domestic shipment.*—Unless otherwise specified in the contract or order, cellulose acetate butyrate shall be packed for domestic shipment in wood-cleated-fiberboard boxes, wood-cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes complying with the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately

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200 pounds; of fiberboard boxes, approximately 70 pounds. (See par. H–2.)

G–1b. *For overseas shipment.*—Unless otherwise specified in the contract or order, cellulose acetate butyrate shall be packed for overseas shipment in cleated plywood boxes, nailed wood boxes, wirebound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes complying with the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Shipping containers shall be lined with a sealed waterproof bag, or its equivalent, complying with the requirements of U. S. Army Specification 19–8. The seams and closures shall be sealed with water-resistant adhesive complying with the requirements of Joint Army-Navy Specification JAN–P–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds. (See par. H–2.)

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipping containers for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Material covered by this specification is intended for use as a component of protective ointment.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the following features:

a. The title, number, and date of the specification.
b. The type of packing desired. (See par. G–1.)

H–3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–4. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion and with the concurrence of the contractor, may modify, in whole or in part, the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Department of the Army will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

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H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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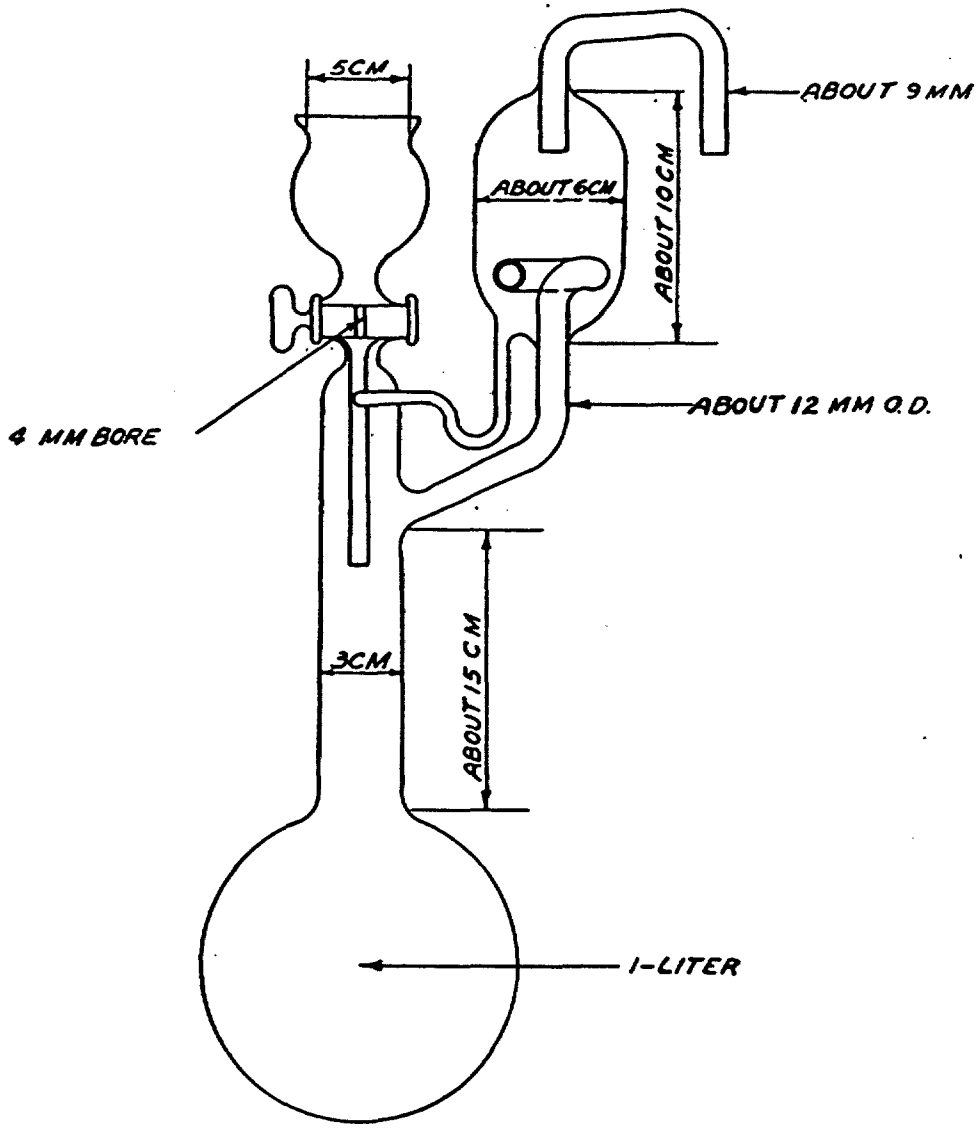


FIGURE 1.—*Pyrex vacuum distilling flask.*

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